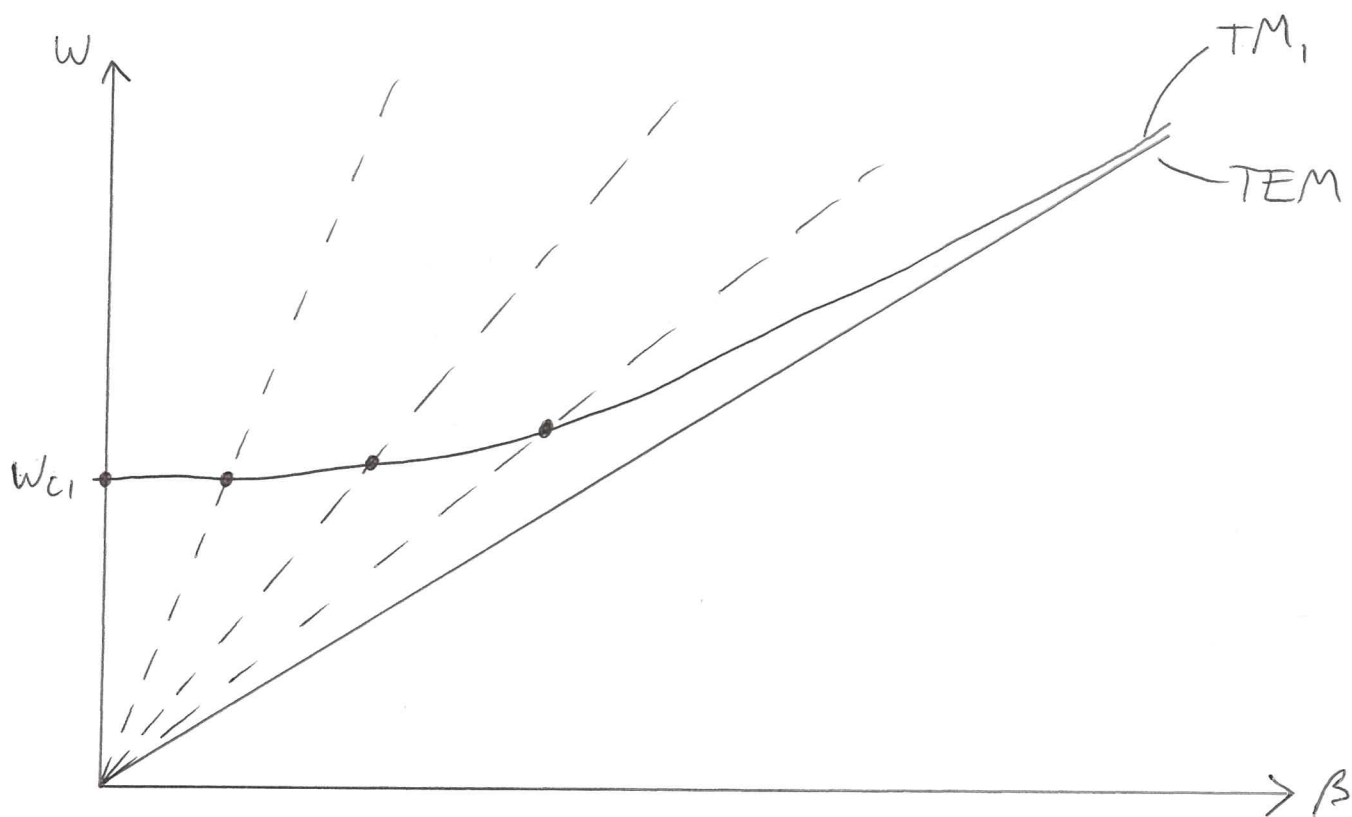


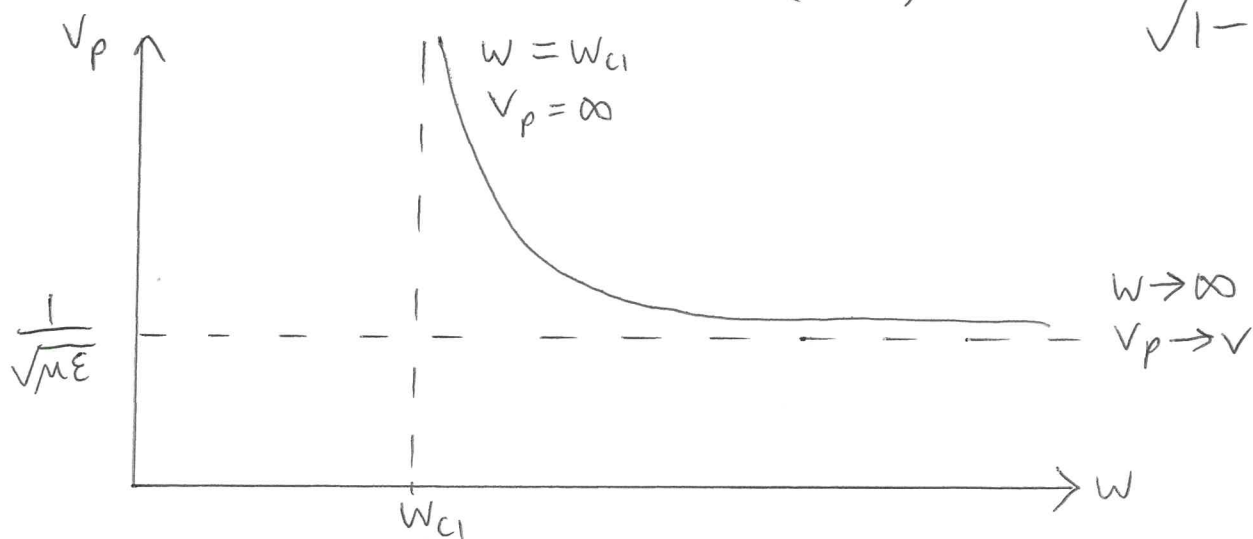
Phase velocity:



$$V_p(\omega) = \frac{\omega}{\beta(\omega)}$$

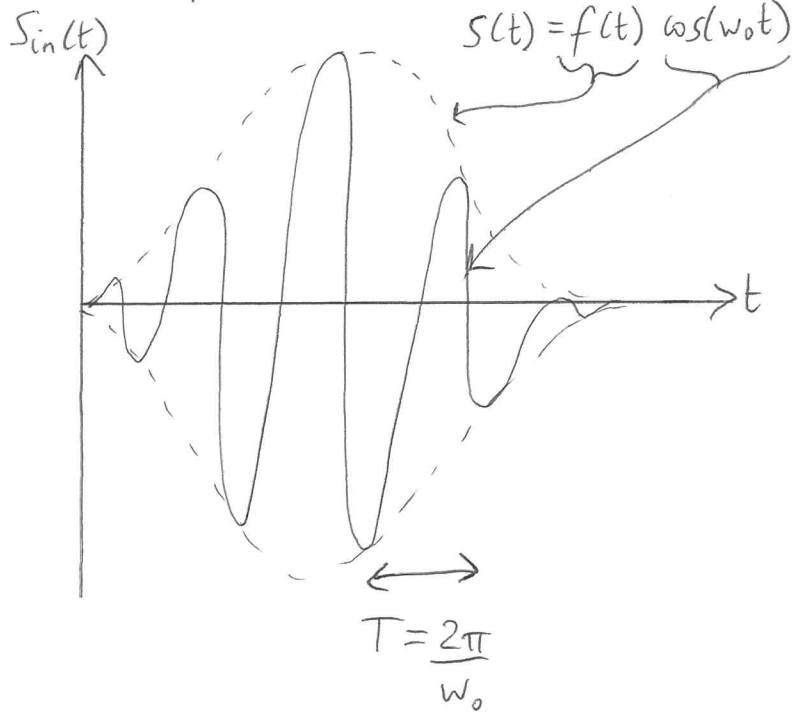
For TEM mode: $\beta(\omega) = \frac{\omega}{v} \quad \therefore v_p = v = \frac{1}{\sqrt{\mu\epsilon}}$

For TM_1 mode: $\beta(\omega) = \frac{\omega}{v} \sqrt{1 - \left(\frac{\omega_{c1}}{\omega}\right)^2} \quad \therefore v_p = \frac{v}{\sqrt{1 - \left(\frac{\omega_{c1}}{\omega}\right)^2}}$



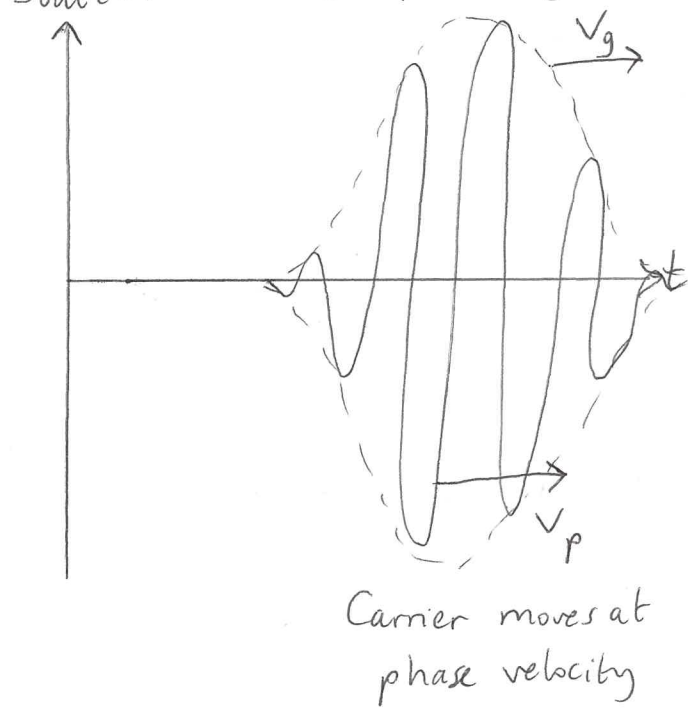
Slide 117)

Group velocity:



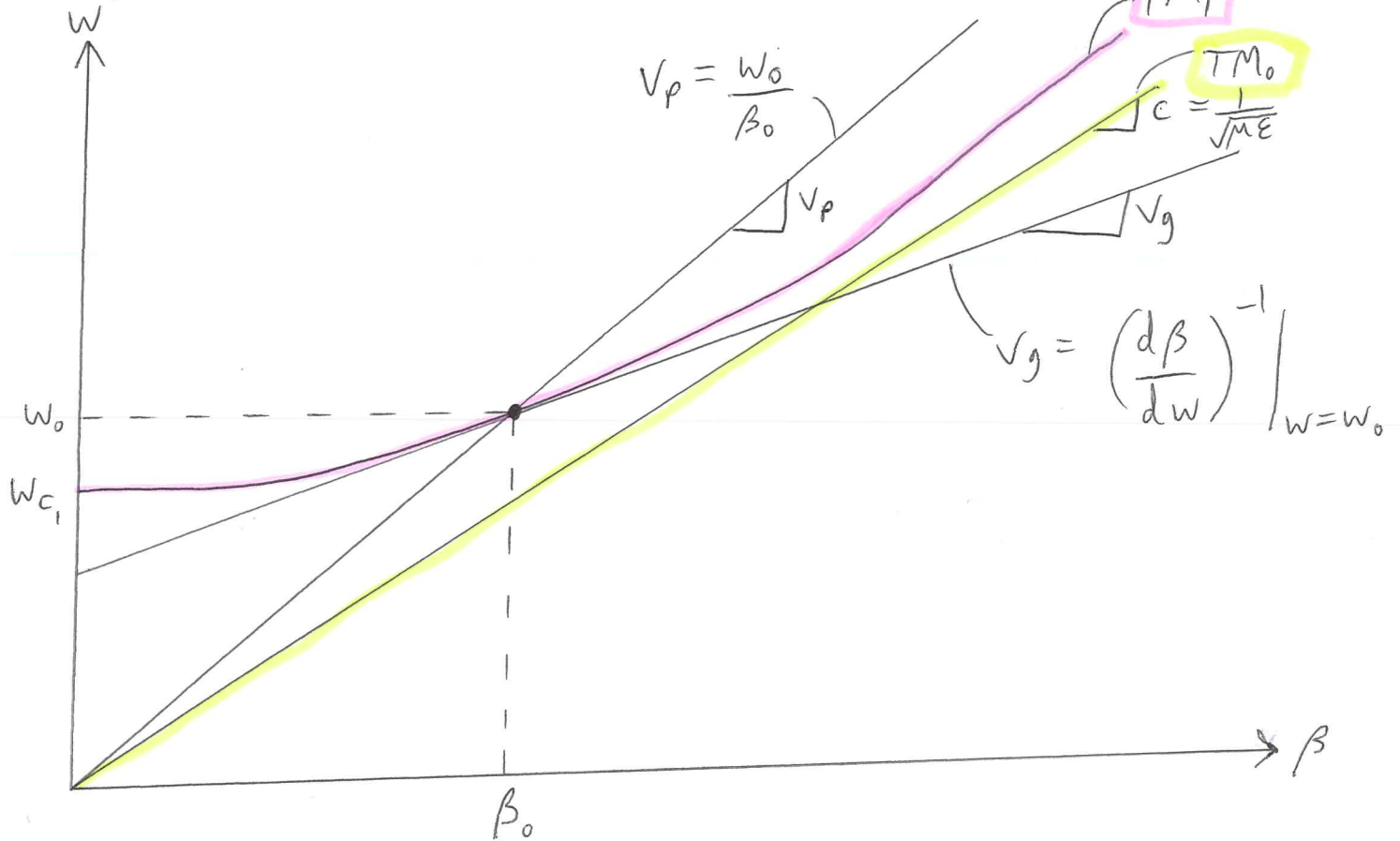
$S_{out}(t)$

Envelope moves at group velocity



Slide 119)

Dispersion relation:



$$V_g \leq c \leq V_p$$

At $w = w_{c1}$: $V_p = \infty$, $V_g = 0$

As $w \rightarrow \infty$: $V_p \rightarrow v = \frac{1}{\sqrt{\mu\epsilon}}$, $V_g \rightarrow v = \frac{1}{\sqrt{\mu\epsilon}}$